

Test Plan Document

Project Name: CampusBookshare

1. Introduction

This test plan outlines the testing strategy for the CampusBookshare application. The goal is to ensure that all features function as expected and meet the requirements. The testing process will include unit testing, system testing, and user acceptance testing.

2. Features to Be Tested

- Backend Features
 - User registration, login, and authentication.
 - CRUD operations for resource listings.
 - Messaging system for user communication.
 - Reviews and ratings for users.
 - Search functionality based on resource category, keywords, and location.
- Frontend Features
 - Responsive web design for all pages.
 - User dashboard to manage listings, messages, and reviews.
 - Messaging interface with active conversations and user list.
 - Listing details page with owner reviews and ratings.
 - User profile view and edit functionality.
- Security Features
 - SQL injection protection.
 - Session management using Flask-Login.
 - Validation for all user inputs (e.g., email format, required fields).

3. Types of Testing

1. Unit Testing

Focuses on individual components of the application, such as routes and database models.

Example: Testing the /register route to ensure proper validation and user creation.

2. System Testing

Verifies that the entire application works as a whole.

Example: Testing the messaging system to ensure users can send and receive messages.

3. User Acceptance Testing (UAT)

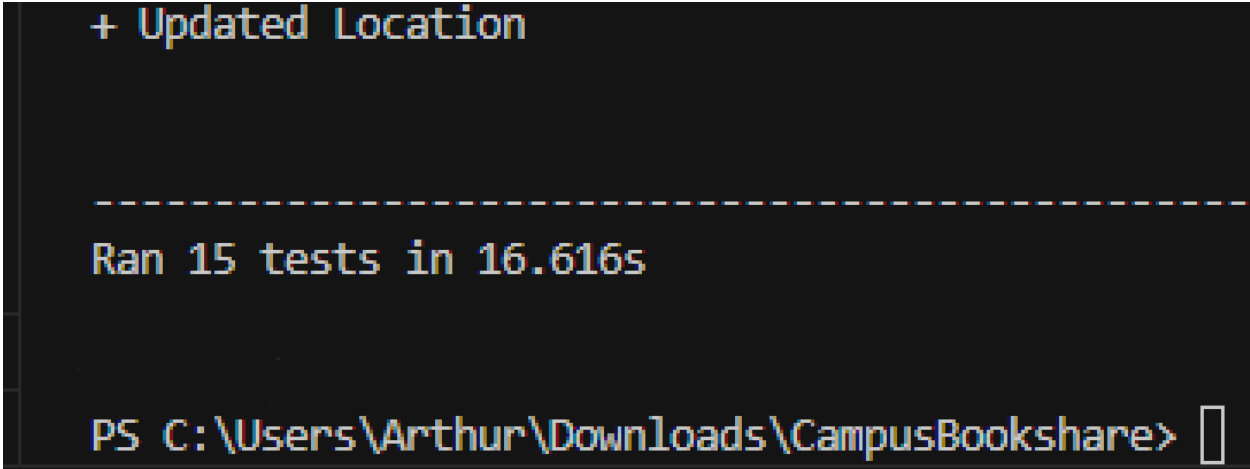
Ensures the application meets user expectations.

Example: Testing the user dashboard to confirm that users can manage their listings and reviews.

4. Testing Tools

Unit Testing: Python's unittest module.

I ran the tests using the terminal itself to ensure the tests ran past clean.



Halfway through I realized I didn’t need so many, so here's a handful of them with an extra Unit-Level Manual test

5. Test Results

Unit-Level Manual Tests					
Test Case	Feature	Steps	Expected Outcome	Pass/Fail	Tester
Test Case 1	User Registration	Navigate to /register. Enter valid name, email, password, and location. Submit the form.	User is successfully registered and redirected to the dashboard.	Pass	Arthur Zhang
Test Case 2	Invalid Registration	Navigate to /register. Leave fields blank or enter invalid email. Submit the form.	Error message is displayed: "All fields are required."	Pass	Arthur Zhang
Test Case 3	User Login	Navigate to /login. Enter valid email and password. Submit the form.	User is successfully logged in and redirected to the dashboard.	Pass	Arthur Zhang
Test Case 4	Invalid Login	Navigate to /login. Enter invalid email or password. Submit the form.	Error message is displayed: "Login failed. Check your email and password."	Pass	Arthur Zhang
Test Case 5	Create a Listing	Log in as a user. Navigate to /create_listing. Enter valid listing details and submit.	Listing is successfully created and appears on the dashboard.	Pass	Arthur Zhang
Test Case 6	Invalid Review Submission	Log in as a user. Navigate to another user's profile. Submit a review with an invalid rating (e.g., 10).	Error message is displayed: "Rating must be between 1 and 5."	Pass	Arthur Zhang

System-Level Tests					
Test Case	Workflow	Steps	Expected Outcome	Pass/Fail	Tester
Test Case 7	Full Workflow for Listing Creation	Register a new user. Log in. Create a listing. Verify the listing appears on the homepage.	Listing is successfully created and visible on the homepage.	Pass	Arthur Zhang
Test Case 8	Full Messaging Workflow	Log in as User A. Send a message to User B. Log in as User B and reply.	Messages are successfully exchanged between User A and User B.	Pass	Arthur Zhang
Test Case 9	Search Functionality	Log in as a user. Search for a keyword in the search bar. Verify results.	Relevant listings and users are displayed based on the search query.	Pass	Arthur Zhang

User Acceptance Tests

Test Case	Scenario	Steps	Expected Outcome	Pass/Fail	Tester
Test Case 10	User Browsing Listings	Log in as a user. Browse listings on the homepage. Click on a listing to view details.	User can browse listings and view details without errors.	Pass	Arthur Zhang
Test Case 11	User Profile Update	Log in as a user. Navigate to /profile. Update name, email, and location.	Profile is updated successfully, and changes are reflected on the profile page.	Pass	Arthur Zhang

6. Conclusion

This test plan ensures that all critical features of the CampusBookshare application are thoroughly tested. The results will help identify and resolve any issues, ensuring a robust and user-friendly application.

Overall minimal issues, although it might be only explained by a not-so-thoroughly executed test program. I'm submitting this late and amidst finals, so I do hope you forgive the lack of quality and screenshots. If you open the folder the tests.py should be in there and you can see on your own.